Tuesday Notes 2/10

Theorem 5: If $m \in \mathbb{N}$, then $m < m+1$

Theorem 6: (Transitivity) Suppose $x, y, z \in \mathbb{N}$ & $x < y$, $x < z$ then $x < z$.

Proof. By def 5, $y = x + b$ and $z = y + c$ for some $b, c \in \mathbb{N}^+$

Thus $z = y + c = (x+b) + c \overset{\text{Ch 1}}{=} x + (b+c)$. By chapter 1, $b + c \neq 0$

by def

$b + c = \sigma^c(b) = \emptyset$

Sidetrack:

if $c \neq 0$, then $c \in \mathbb{N}^+$

Thus $c = \sigma(a)$ for some a in $\mathbb{N}$

Thus $\sigma^c(b) = \sigma^{\sigma(a)}(b) = \sigma(\sigma^a(b)) = \emptyset$ by Ax 2

$c = \emptyset$ Axiom 5

$id(b) = b = \emptyset$

SO, $b + c \in \mathbb{N}^+$ Thus $x < z$ ✓

Lemma 7. If $m \in \mathbb{N}$ then $m = \emptyset$ or $\emptyset < m$

(Exactly one of the following holds)

Lemma 8: (Weak trichotomy)

If $m, n \in \mathbb{N}$, then at least one of the following holds $m \leq n$, $m = n$, $m > n$

Thm. Other Paper

Thm 9) $<$ is a strict linear order on $\mathbb{N}$

Proof: We have shown $<$ is transitive (thm 6)
By Lemma 8, we have weak trichotomy if we can show $x \not< x$ for any $n \in \mathbb{N}$, then we are done by Exercise 2!

By contradiction: Suppose $x < x$

Then $x = x + b$ for some $b \in \mathbb{N}^+$

~~Otherwise~~ In Chap 1: $x = x + 0$, so we have $x + 0 = x + b$. by cancellation (ch 1) $0 = b$ meaning $0 \in \mathbb{N}^+$, but 0 isn't in the $\mathbb{N}^+$

We see $x \not< x$ ✓

DEF: $x \leq y$ if $x < y$ or $x = y$

Thm 10. $0 \leq n \quad \forall n \in \mathbb{N}$

Thm 11. If $m, n \in \mathbb{N}$ then $m \leq n$ if there exists $a \in \mathbb{N}$ with $m + a = n$

Proof Sketch: If $m < n$ then $a \in \mathbb{N}^+$ by def of $<$
If $m = n$, then $m = m + 0 = n$ or $a = 0$ in other words

2.3 Basic Properties of the order on $\mathbb{N}$

Thm 12 $n \neq 0 \iff n > 0$

DEF: $m > n$ if $n < m$ for all $\in \mathbb{N}$

Cor 13. $n \in \mathbb{N}^+ \iff n > 0$

Thm 14. Let $n \in \mathbb{N}$. There is no x such that $n < x < n+1$

Proof: Suppose $n < x$ & $x < n+1$, Thus $x = n + b$ for some $b \in \mathbb{N}^+$ and $x + c = n+1$ for $c \in \mathbb{N}^+$.

Therefore, $x = n + b = n + \sigma(d) = n + d + 1 = n + 1 + d = x + c + d$

$c + d \in \mathbb{N}^+$, so $x < x$ contradicting trichotomy. We conclude no $\mathbb{N}$ between n and $n+1$ ✓